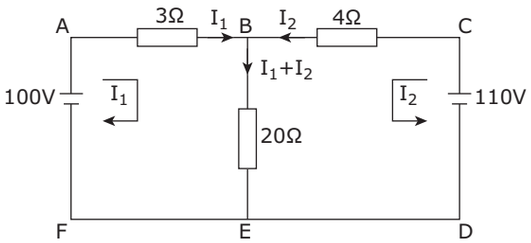
IV. Explain in about a Page. **Each question carries ten marks.**

1. Using Kirchhoff's law, find the value of current flow through all resistors in the given circuit



2. In a series circuit, two resistors 50Ω and 100Ω are connected with 300 V supply. Prove that the sum of the voltage drops across the resistors is equal to the EMF applied in the circuit.
3. In a parallel circuit, two resistors 100 Ω and 150Ω are connected with 240V supply. Prove that the sum of branch currents is equal to the Total current of the circuit.
4. List out the functions of a capacitor.

Answers

1) **b** 2) **b** 3) **a** 4) **d** 5) **c**

6) **d** 7) **b** 8) **c** 9) **b** 10) **c**

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